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**To be cited as:** 10.1002/anie.201708489

**Link to VoR:** <https://doi.org/10.1002/anie.201708489>

# Heavy-atom tunneling calculations in thirteen organic reactions: tunneling contributes to the rate, and Bell's formula closely approximates multidimensional tunneling at $\geq 250$ K

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In honor of Roald Hoffmann's 80<sup>th</sup> birthday.

**Abstract:** Multidimensional tunneling calculations are carried out for 13 reactions, to test the scope of heavy-atom tunneling in organic chemistry, and to check the accuracy of one-dimensional tunneling models. The reactions include pericyclic, cycloaromatization, radical cyclization and ring opening, and  $S_N2$ . When compared at the temperatures that give the same effective rate constant of  $3 \times 10^{-5} \text{ s}^{-1}$ , tunneling accounts for 25 – 95% of the rate in 8 of the 13 reactions. Values of transmission coefficients predicted by Bell's formula,  $\kappa_{\text{Bell}}$ , agree well with multidimensional tunneling (canonical variational transition state theory with small curvature tunneling),  $\kappa_{\text{SCT}}$ . Mean unsigned deviations of  $\kappa_{\text{Bell}}$  vs  $\kappa_{\text{SCT}}$  are 0.08, 0.04, 0.02 at 250, 300 and 400 K. This suggests that  $\kappa_{\text{Bell}}$  is a useful first choice for predicting transmission coefficients in heavy-atom tunnelling.

Heavy-atom tunneling has been found experimentally or predicted computationally in a number of reactions,<sup>1,2</sup> even at ordinary reaction temperatures.<sup>1kl</sup> In several of these studies, the authors suggest that heavy-atom tunneling may be broadly important, but no computational survey of heavy-atom tunneling in common reaction types has yet appeared. Here we report such a study carried out for 13 reactions (Scheme 1), using well validated multidimensional tunneling methods,<sup>3</sup> and comparing the results with the 1-dimensional tunneling models of Wigner,<sup>4</sup> Bell<sup>5,6</sup> and Truhlar.<sup>7</sup> For the molecules studied here, the Bell and Truhlar predictions are identical, and they agree well with multidimensional tunneling calculations in the temperature range 250–450 K, at which the reactions are normally carried out.

The rate constant in transition state theory (TST) is  $k = \kappa(RT/h)\exp(-\Delta G^\ddagger/RT)$ , where  $\kappa$ , the transmission coefficient, is the quantum correction for motion along the reaction coordinate,  $R$  is the gas constant,  $h$  is Planck's constant,  $T$  is temperature, and  $\Delta G^\ddagger$  is the free energy of activation.  $\kappa$  includes the effect of forward transmission through the barrier and backward reflection above it.  $\kappa$  is the ratio of the tunnel-corrected rate to the classical TST rate;  $\kappa = 1$  if there is no tunneling and  $\kappa > 1$  if tunneling contributes to the rate. We take  $(\kappa - 1)/\kappa$  as a convenient

measure of the fraction of reaction due to tunneling.

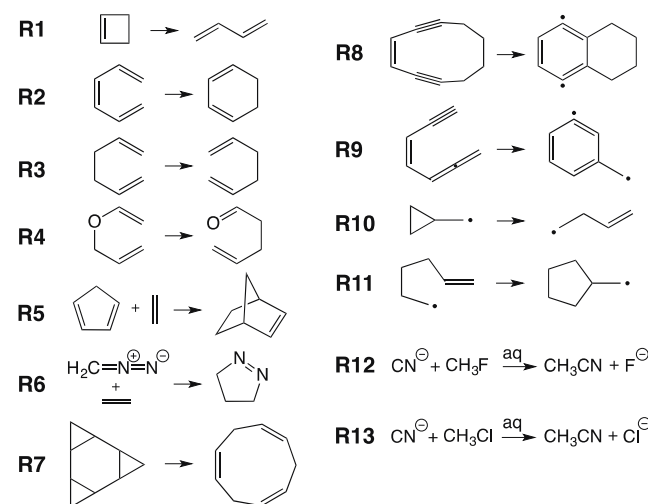
The 1-dimensional tunneling methods considered here are based on a quadratic potential. Wigner examined the quantum correction to the TST rate constant in the limit  $\kappa \approx 1$ . Neglecting terms of higher order than  $h^2$ , he obtained

$$\kappa_W = 1 + \frac{u^2}{24}, \quad (1)$$

where  $u = h\nu^\ddagger/RT$  and  $i\nu^\ddagger$  is the imaginary frequency at the TS. Using a truncated parabolic barrier model, Bell<sup>5,6</sup> and Truhlar<sup>7</sup> derived complementary formulas for  $\kappa$  (Supporting Information, Eqs S1 and S2, respectively). At high temperature, the leading term for both is

$$\kappa_{\text{Bell}} = (u/2)/\sin(u/2), \quad u < 2\pi, \quad (2)$$

the well-known result for an infinite parabolic barrier.<sup>5,7</sup> The additional terms of Eqs S1 and S2 are negligible for R1–R13 except at very low temperature.



**Scheme 1.** Reactions examined in this study.

Scheme 1 includes pericyclic reactions (electrocyclic, sigmatropic, cycloadditions), cycloaromatizations, radical rearrangements, and  $S_N2$  reactions with continuum water solvation. All have been studied computationally in different contexts,<sup>8</sup> some involving tunneling.<sup>1h,1i</sup> Tunneling has not been reported in R12 and R13, but Truhlar and co-workers have studied tunneling in the gas phase  $S_N2$  reactions  $\text{Cl}^-(\text{H}_2\text{O}) + \text{CH}_3\text{Cl} \rightarrow \text{ClCH}_3 + \text{Cl}^-(\text{H}_2\text{O})$ ,<sup>9a</sup>  $\text{CH}_3\text{NH}_3^+\text{Cl}^- \rightarrow \text{CH}_3\text{Cl} + \text{NH}_3$  and  $\text{CH}_3\text{Cl}(\text{H}_2\text{O}) + \text{NH}_3(\text{H}_2\text{O}) \rightarrow \text{CH}_3\text{NH}_3^+\text{Cl}^-(\text{H}_2\text{O})_2$ .<sup>9b</sup>

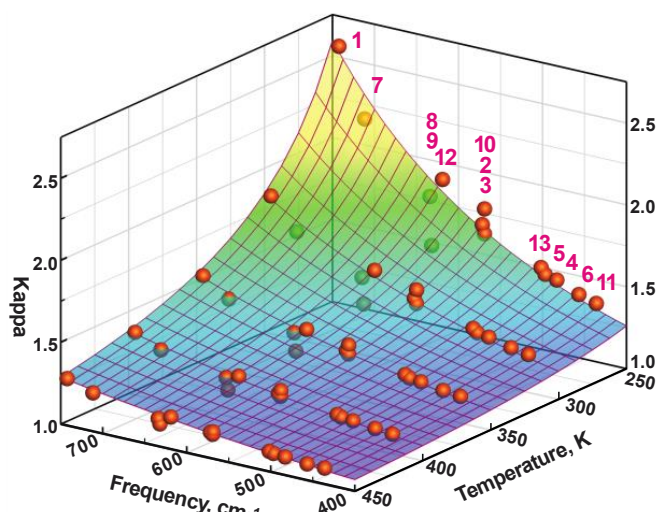
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Multidimensional tunneling calculations on R1-R13 were carried out with Polyrate<sup>3,10</sup> using the small curvature tunneling (SCT) approximation, with the Gaussrate<sup>11</sup> interface to Gaussian 09.<sup>12</sup> Reaction paths were computed by DFT (Supporting Information). The effect of tunneling on the rate constant is given by  $k_{\text{CVT+SCT}} = \kappa_{\text{SCT}} k_{\text{CVT}}$ , where  $k_{\text{CVT}}$  is the rate constant excluding tunneling computed by canonical variational transition state theory, or CVT,<sup>13</sup>  $\kappa_{\text{SCT}}$  is the SCT transmission coefficient, and  $k_{\text{CVT+SCT}}$  is the rate constant including SCT tunneling. The time-consuming step is the construction of the effective barrier for tunneling, which is the vibrationally adiabatic ground state curve  $V_A^G$ . It consists of the intrinsic reaction coordinate (IRC) plus the vibrational zero point energy (ZPE) at each point on the path.

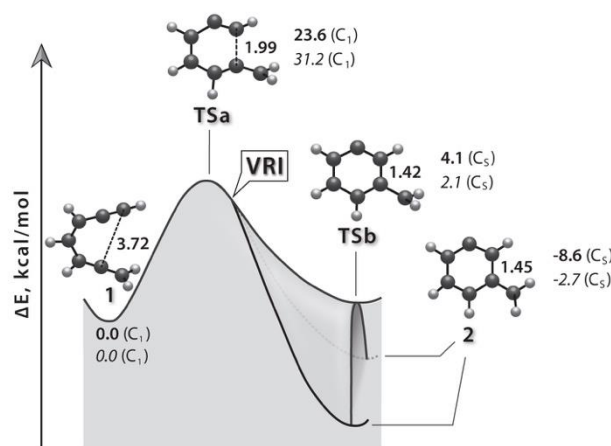
The results are summarized in Figure 1 and Table 1. In Figure 1,  $\kappa_{\text{SCT}}$  values computed by Polyrate at 250 – 450 K are superimposed on a  $\kappa_{\text{Bell}}$  surface. Red numbers refer to reactions R1–R13. Table S1 in the Supporting Information gives the values of  $\kappa_{\text{SCT}}$  in Figure 1, and Tables S2 and S3 give unsigned errors of  $\kappa_W$  and  $\kappa_{\text{Bell}}$  vs  $\kappa_{\text{SCT}}$ . The mean unsigned errors (MUEs) for  $\kappa_W$  vs  $\kappa_{\text{SCT}}$  are 0.25, 0.11, 0.04 at 250, 300 and 400 K. For  $\kappa_{\text{Bell}}$  vs  $\kappa_{\text{SCT}}$ , MUEs are 0.08, 0.04, 0.02 at 250, 300 and 400 K; excluding R9, MUEs for  $\kappa_{\text{Bell}}$  are 0.06, 0.03 and 0.01.



**Figure 1.** Comparison of  $\kappa_{\text{Bell}}$  (surface) to  $\kappa_{\text{SCT}}$ , the Polyrate SCT transmission coefficient (spheres).  $\kappa_{\text{SCT}}$  points below the surface appear submerged. Red numbers refer to reactions R1 – R13. Mean unsigned errors (MUEs) for  $\kappa_{\text{Bell}}$  vs  $\kappa_{\text{SCT}}$  are 0.08, 0.04, 0.02 at 250, 300 and 400 K. Excluding R9, MUEs are 0.06, 0.03 and 0.01.

R9, and to a lesser degree R8, have the largest deviations of  $\kappa_{\text{Bell}}$  vs  $\kappa_{\text{SCT}}$ . In these reactions, the unrestricted Kohn-Sham self-consistent field (UKS-SCF) solutions switch from closed to open shell near the TS. This gives a discontinuous ZPE along the path, which we replaced by a continuous ZPE derived from scaled 10,10-CASSCF frequencies while retaining the DFT-based IRC to obtain a continuous  $V_A^G$  curve with satisfactory energetics.<sup>11</sup> In addition, the IRC for R9 bifurcates on the product side<sup>14</sup> (Figure 2) as the allenic  $\text{CH}_2$  rotates clockwise vs counterclockwise into the plane of the ring to produce isomeric

diradicals **2**. The valley-ridge inflection (VRI) point, where the initial IRC valley changes to a ridge, occurs  $\sim 1$  kcal/mol below **TSa**, within the tunneling region. The combined effect of the SCF switch and VRI, both near the TS, leads to a large deviation of  $\kappa_{\text{Bell}}$  from  $\kappa_{\text{SCT}}$  in R9.



**Figure 2.** Bifurcating PES for the Myers-Saito reaction, R9. Stationary points were optimized with 10,10-CASSCF/6-31G(d). Energies in italic type include CASSCF vibrational zero-point energies scaled by 0.91. Energies in bold type were obtained by single point calculations of the stationary points with modified BLYP/6-31G(d).<sup>11</sup> The valley-ridge inflection (VRI) located in the tunnelling region splits the IRC into two branches that lead to isomeric diradicals **2** that differ in the position of the H atoms of the  $\text{CH}_2$  group.

In Table 1, we compare the reactions at temperatures that give a common first order rate constant of  $3 \times 10^{-5} \text{ s}^{-1}$  (pseudo-first order for bimolecular reactions). This rate constant was chosen to allow most of the reactions to be compared at a rate and temperature that can be encountered in running these reactions. (At a fixed rate constant, however, not all temperatures are typical.) R10 cannot be included because its minimum tunneling rate is predicted to be  $2 \times 10^{-2} \text{ s}^{-1}$ .<sup>1h</sup> Eight reactions, including R10, have tunneling fractions  $\geq 25\%$ , at temperatures up to 420 K.

The lowest temperature in Table 1 occurs for R11 at 104 K, below the normally applicable range of Eq 1 and Eq 2 to reactions R1-R13. With an additional term in the Bell series<sup>5,6</sup> (Eq S1),  $\kappa_{\text{Bell}}$  is fairly close to  $\kappa_{\text{SCT}}$  even at 104 K.

The results in Figure 1 and Table 1 show that Eq 2 is largely successful at mimicking  $\kappa_{\text{SCT}}$  for R1-R13 over 250 – 450 K. The results further suggest that tunneling makes substantial contributions to the rates of electrocyclic (R1 and R2) and  $\text{S}_{\text{N}}2$  (R12 and R13) reactions, some cycloaromatizations (R8 and R9), and radical cyclization (R11) and ring opening (R7, with R10 previously characterized).<sup>1h</sup> Tunneling contributes a smaller amount in cycloadditions R5, R6 and [3,3]-shifts R3 and R4, although Borden computed dominant tunneling in semibullvalene [3,3]-shifts, which have a much lower barrier.<sup>1g</sup>

**Table 1.** Percent tunneling and transmission coefficients at temperatures that give a common rate constant of  $3 \times 10^{-5} \text{ s}^{-1}$ .<sup>[a]</sup>

|     | $\Delta G^\ddagger$<br>0 K <sup>[b]</sup> | $\Delta G^\circ$<br>0 K <sup>[b]</sup> | $\nu^\ddagger$ , <sup>[c]</sup><br>$\text{cm}^{-1}$ | T, K           | % Tun <sup>[d]</sup> | $\kappa_{\text{SCT}}$ <sup>[e]</sup> | $\kappa_{\text{Bell}}$ <sup>[f]</sup> | $\kappa_{\text{W}}$ <sup>[g]</sup> |
|-----|-------------------------------------------|----------------------------------------|-----------------------------------------------------|----------------|----------------------|--------------------------------------|---------------------------------------|------------------------------------|
| R1  | 33.9                                      | -9.3                                   | 742                                                 | 420            | 25 %                 | 1.33                                 | 1.33                                  | 1.27                               |
| R2  | 20.5                                      | -22.3                                  | 572                                                 | 272            | 35 %                 | 1.54                                 | 1.52                                  | 1.38                               |
| R3  | 33.5                                      | 0.0                                    | 569                                                 | 456            | 13 %                 | 1.15                                 | 1.15                                  | 1.13                               |
| R4  | 26.4                                      | -18.1                                  | 483                                                 | 349            | 17 %                 | 1.20                                 | 1.19                                  | 1.17                               |
| R5  | 22.3                                      | -18.4                                  | 497                                                 | 491            | 9 %                  | 1.10                                 | 1.09                                  | 1.09                               |
| R6  | 16.6                                      | -30.6                                  | 457                                                 | 314            | 19 %                 | 1.22                                 | 1.21                                  | 1.18                               |
| R7  | 21.8                                      | -20.8                                  | 712                                                 | 269            | 50 %                 | 1.92                                 | 2.02                                  | 1.60                               |
| R8  | 24.8                                      | 8.0                                    | 634                                                 | 324            | 28 %                 | 1.38                                 | 1.43                                  | 1.33                               |
| R9  | 24.0                                      | -8.2                                   | 632 <sup>[h]</sup>                                  | 330            | 20 %                 | 1.25                                 | 1.43 <sup>[h]</sup>                   | 1.32                               |
| R10 | 7.6                                       | -3.1                                   | 569                                                 | <sup>[i]</sup> | —                    | —                                    | —                                     |                                    |
| R11 | 8.4                                       | -12.9                                  | 436                                                 | 104            | 95 %                 | 18.6                                 | 19.6 <sup>[j]</sup>                   | 2.52                               |
| R12 | 20.2                                      | -24.6                                  | 619                                                 | 340            | 29 %                 | 1.38                                 | 1.36                                  | 1.29                               |
| R13 | 12.2                                      | -42.5                                  | 502                                                 | 212            | 45 %                 | 1.57                                 | 1.72                                  | 1.48                               |

[a] Pseudo-first order for bimolecular reactions. [b] kcal/mol. [c]  $i\nu^\ddagger$  = imaginary frequency at the TS. [d]  $(\kappa_{\text{SCT}} - 1)/\kappa_{\text{SCT}}$ . [e] Polyrates small curvature tunneling transmission coefficient. [f] From Eq 2 at the given temperature. [g] From Eq 1. [h] R9 has a valley-ridge inflection near the TS that affects  $\nu^\ddagger$ . [i] The minimum rate (100% tunneling at 20 K) is computed to be  $0.02 \text{ s}^{-1}$  (Ref 1h). [j] An additional term is needed for  $\kappa_{\text{Bell}}$ . The Truhlar Eq S2 gives the same result.

Eq 2 succeeds because heavy-atom tunneling above 250 K approximately fulfils the assumptions in the derivation. (1) Most reactions R1-R13 have high barriers that require high temperatures where  $u \ll 2\pi$  and Eq 2 is most accurate. (2) With larger mass and smaller tunneling probabilities, the heavy-atom tunneling region is near the TS, where the barrier width is narrow. This is approximately the quadratic region of the surface as assumed in the 1-D methods. (3) Reaction path curvature (in the 3N-6 dimensional space of vibrational modes) is small in the tunneling regions of R1-R13. That is, the path direction remains nearly constant and parallel to the transition vector, making the tunneling path effectively one-dimensional. This is important because a curved reaction path presents a corner-cutting opportunity for shortening the tunneling path and increasing  $\kappa$  beyond the value predicted for a 1-D potential.<sup>3,15,16</sup>

Eq 2 has long been recognized as a useful correction for heavy-atom KIEs. Singleton and coworkers<sup>17</sup> have applied it routinely in a number of reactions, including [3,3]-shifts, ene reactions of singlet oxygen, cycloaromatization, solvolysis, and cycloadditions. In a study of Roush allylboration, Veticatt and Singleton<sup>1k</sup> chose the more accurate CVT/SCT method and obtained  $\kappa_{\text{SCT}} = 1.36$  at 195 K. From their published TS geometry we computed  $\nu^\ddagger = 338 \text{ cm}^{-1}$  and  $\kappa_{\text{Bell}} = 1.32$ , close to  $\kappa_{\text{SCT}}$  even at 195 K. In semibullvalene tunneling,<sup>1g</sup> Borden

reported  $\kappa_{\text{SCT}} = 1.25$  at 150 K and  $\kappa_{\text{SCT}} = 1.79$  and 100 K. With Borden's  $\nu^\ddagger = 225.9 \text{ cm}^{-1}$ , Eq 2 gives  $\kappa_{\text{Bell}} = 1.23$  at 150 K, close to  $\kappa_{\text{SCT}}$ , but deviates at 100 K with  $\kappa_{\text{Bell}} = 1.63$ . In allylboration and semibullvalene cases, it is encouraging that Eq 2 can be useful at temperatures well below 250 K.

A reviewer asked how well Eq 2 works with cyclobutadiene automerization.<sup>18</sup> At 250 K with  $\nu^\ddagger = 1929 \text{ cm}^{-1}$  (Dewar, et al.<sup>19</sup>) Eq 2 with additional terms gives  $\kappa = 70 - 175$  depending on the barrier (4 – 5 kcal/mol); the Truhlar  $\kappa$  is 2.5% higher. This is consistent with Carpenter's finding that tunneling dominates the rate.<sup>18</sup>

In summary, we have studied several common reaction types to assess the importance of heavy-atom tunneling in organic reactions. The results are consistent with the hypothesis that heavy-atom tunneling is widespread. The success of Eq 2 vs multidimensional tunneling calculations suggests it is a useful first choice for predicting transmission coefficients in heavy-atom tunnelling.

## Acknowledgements

E.M.G. and C.V.C. acknowledge support from the Donors of the American Chemical Society Petroleum Research Fund, Grant No. 54244-UR4 and the City University of New York PSC-CUNY Research Award Program. Computational support was provided by the Extreme Science and Engineering Discovery Environment (XSEDE), which is supported by National Science Foundation Grant No. ACI01053575. C.D. thanks the National Science Foundation, grant CHE-1465040, for financial support and XSEDE grant TG-CHE090070 for computational support.

**Keywords:** heavy-atom tunneling • transmission coefficients • Bell's formula • Wigner's formula

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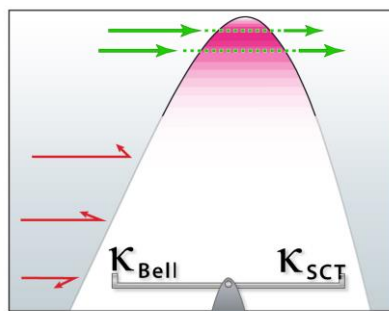
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